Assignment No-2

Title:- Building an Expert System using Rule-based System.

Objective: Develop an Expert System that provides simple decision-making

Part 2A:

Title: Expert System Using If-else Rules.

Problem statement:

Develop a simple Expert System that accepts marks obtained by a student in five different subjects. The system calculates the total marks and percentages of the student. Based on the calculated percentage, the system displays the appropriate class using 'if...else' conditions

The grading rules are:

Below 40% – "fail"
40% to below 65% – "II class"
65% to below 75% – "I class"
75% and above – "Distinction"

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Theory:

An Expert System is an Artificial Intelligence system that used stored knowledge and rules to make decisions similar to a human expert. A rule based Expert System generally consists of facts, rules, and an inference mechanism.

In this problem, the marks entered by the user act as the input facts. The percentage is calculated from these marks, and "if-else" statements are used as rules to determine the student's class.

The system accept marks for five subjects. Since each subject is assumed to have a maximum of 100 marks, the maximum total marks are 500. The percentage is calculated using.

"Percentage = (Total marks / 500) x 100)"

After calculating the percentage, the system checks the conditions in sequence. If the percentage is below 40, the student is considered to have failed. if it is between 40 and 65, the student get II class. A percentage from 65 to below 75 results in I class, while 75 or above results in Distinction.

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Algorithm.

1. Start the program
2. Enter the marks obtained in five subjects.
3. Calculate the total marks
4. Calculate the percentage
5. If percentage is below 40, display "Fail"
6. Else if percentage is below 65, Display "II class"
7. Else if percentage is below 75, Display "I class"
8. Else display "Distinction"
9. Display the total marks and percentage
10. Stop the program.

Explanation of Rules:

The decision-making part of the program uses four rules:

"Rule 1" If the percentage is less than 40% the results is "fail".

"Rule 2" If the percentage is 65% or more but less than 65%, the reowlts is "II class"

"Rule 3" If the percentage is 65% or more but less than 75%, the result is "I class"

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Rule 4: If the percentage is 75+ or more, the result is "Distinction".

The 'if', 'elif', and 'else' conducts act as the decision-making mechanism of the Expert System. Only one rule is selected for the final result.

Result:

Thus, a simple rule-based Expert System was successfully developed using Python 'if-else' statements accepts marks for five subjects. calculates the student's percentage, and automatically determines whether the student has failed, obtained II class, I class, or Distinction.

Open *student_grading_system.py

```python
# Python program to calculate percentage and grade

# Input marks for five subjects
sub1 = float(input("Enter marks for Subject 1: "))
sub2 = float(input("Enter marks for Subject 2: "))
sub3 = float(input("Enter marks for Subject 3: "))
sub4 = float(input("Enter marks for Subject 4: "))
sub5 = float(input("Enter marks for Subject 5: "))

# Calculate total and percentage
total = sub1 + sub2 + sub3 + sub4 + sub5
percentage = total / 5

# Display percentage
print("\nPercentage:", percentage, "%")

# Determine grade using if...else
if percentage >= 75:
    print("Grade: Distinction")
elif percentage >= 65:
    print("Grade: I Class")
elif percentage >= 40:
    print("Grade: II Class")
else:
    print("Grade: Fail")

```

```
aids@aids-System-Product-Name: ~
aids@ai...  aids@ai...  aids@ai...  aids@ai...  aids@ai...
aids@aids-System-Product-Name:~$ python3 student_grading_system.py
Enter marks for Subject 1: 85
Enter marks for Subject 2: 78
Enter marks for Subject 3: 98
Enter marks for Subject 4: 94
Enter marks for Subject 5: 76

Percentage: 86.2 %
Grade: Distinction
aids@aids-System-Product-Name:~$
```

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Assignment 2-B

Problem statement: To develop an expert system to suggest a career path and ground on the students interests.

Theory:

Role of knowledge Base and Interference Engine

The knowledge base is the central source of information used by the Expert system. In this program it contains the facts representing student interests and its rules connecting those interests to ~~cover~~ career paths.

The Inference engine is responsible for applying the rules of to the available facts. It checks whether the conditions ~~facts~~ of a rule are satisfied and, if they are, executes the corresponding action.

Thus, the overall decision-making process can be represented as

Student interests → facts → Rule matching → Inference engine → career recommendation.

This process demonstrates how artificial intelligence can be used to make simple decisions without requiring a machine-learning model.

Advantages and Limitations

The developed expert-system is simple, easy to understand and suitable for demonstrating the basic principles of rule-based artificial intelligence. New career paths can be added by creating additional rules. The system also provides consistent recommendations whenever same combination of interests is supplied.

However the system has some limitations. The recommendations depend entirely on the predefined rules. It cannot automatically learn new relationships between subjects and careers. Also, students real career choice depends on many factors such as academic performance, skills, aptitude, which are not considered by this basic system.

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Conclusion

The career path Expert system successfully demonstrates the use of rule based Artificial Intelligence using the Experta Python library. Student interests are stored as facts and predefined rules are used by the inference engine. to suggest suitable career paths. This assignment helped in understanding the practical use of facts, rules, knowledge, Base, and inference engine in an Expert system.

```
from experta import *

class StudentFacts(Fact):
    pass


class CareerExpertSystem(KnowledgeEngine):

    def __init__(self):
        super().__init__()
        self.matched = False

    @Rule(StudentFacts(likes='maths'),
          StudentFacts(likes='programming'))
    def computer(self):
        self.matched = True
        print("Suggested Career Path: Computer Engineering")

    @Rule(StudentFacts(likes='mechanics'),
          StudentFacts(likes='maths'))
    def mechanical(self):
        self.matched = True
        print("Suggested Career Path: Mechanical Engineering")

    @Rule(StudentFacts(likes='anatomy'),
          StudentFacts(likes='pharmacology'))
    def mbbs(self):
        self.matched = True
        print("Suggested Career Path: MBBS")

    @Rule(StudentFacts(likes='circuits'),
          StudentFacts(likes='maths'))
    def electronics(self):
        self.matched = True
        print("Suggested Career Path: Electronics Engineering")

    @Rule(StudentFacts(likes='legal writing'),
          StudentFacts(likes='constitutional law'))
    def law(self):
        self.matched = True
        print("Suggested Career Path: LLB")

    @Rule(StudentFacts(likes='digital media'),
          StudentFacts(likes='advertising'))
    def media(self):
        self.matched = True
        print("Suggested Career Path: Journalism")

    @Rule(StudentFacts(likes='architectural design'),
          StudentFacts(likes='urban planning'))
    def architecture(self):
        self.matched = True
        print("Suggested Career Path: B.Arch")


def main():
    engine = CareerExpertSystem()
    engine.reset()

    print("Welcome to the Career Path Expert System!")

    print("\nAvailable interests:")
    print("maths")
    print("programming")
```

```
    print("mechanics")
    print("anatomy")
    print("pharmacology")
    print("circuits")
    print("legal writing")
    print("constitutional law")
    print("advertising")
    print("digital media")
    print("urban planning")
    print("architectural design")

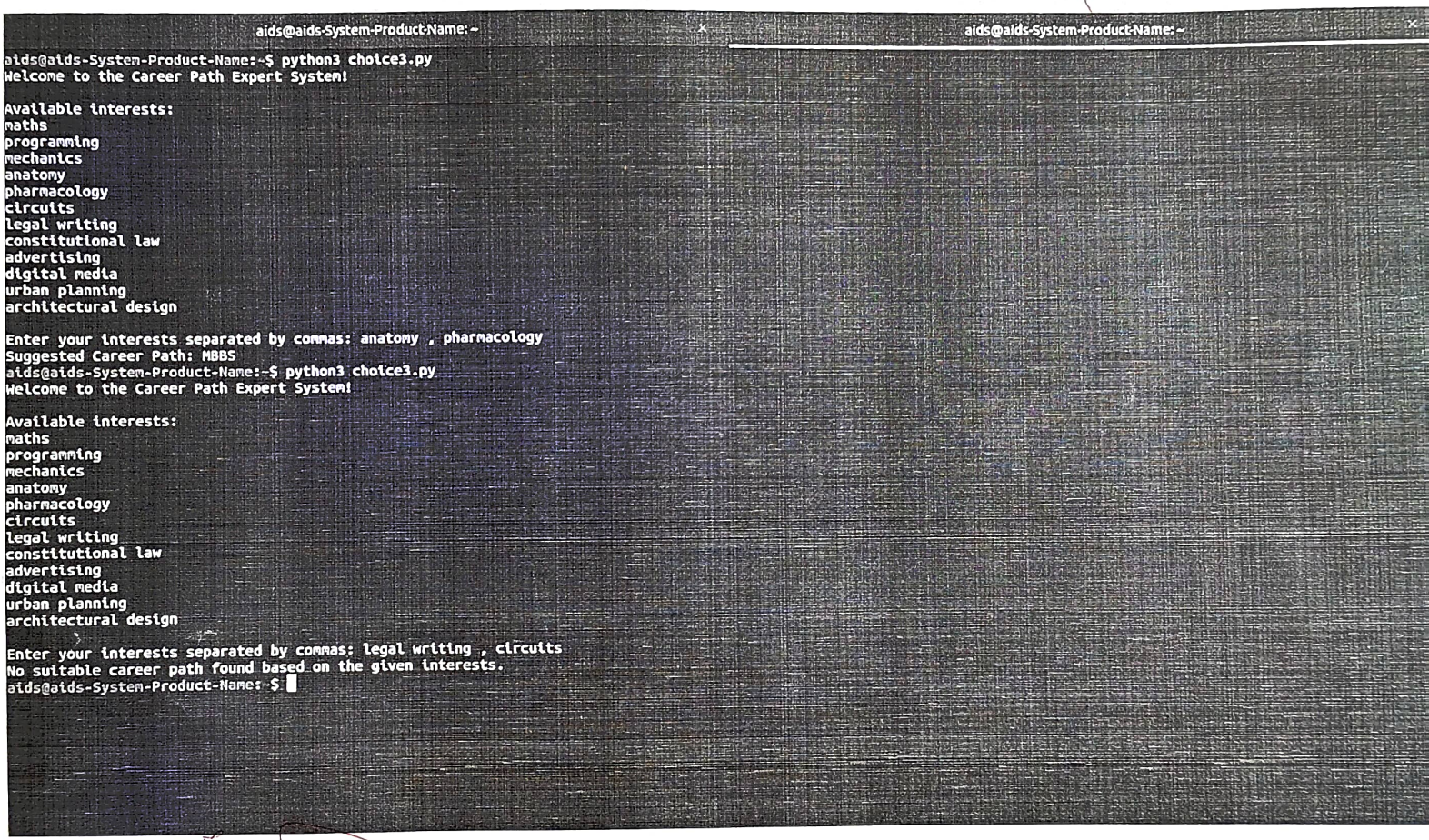
    interests = input("\nEnter your interests separated by commas:
").lower().split(',')

    for interest in interests:
        engine.declare(StudentFacts(likes=interest.strip()))

    engine.run()

    if not engine.matched:
        print("No suitable career path found based on the given interests.")


if __name__ == "__main__":
    main()
```


aids@aids-System-Product-Name:~
aids@aids-System-Product-Name:~
aids@aids-System-Product-Name:~$ python3 choice3.py
Welcome to the Career Path Expert System!
Available interests:
maths
programming
mechanics
anatomy
pharmacology
circuits
legal writing
constitutional law
advertising
digital media
urban planning
architectural design
Enter your interests separated by commas: anatomy , pharmacology
Suggested Career Path: MBBS
aids@aids-System-Product-Name:~$ python3 choice3.py
Welcome to the Career Path Expert System!
Available interests:
maths
programming
mechanics
anatomy
pharmacology
circuits
legal writing
constitutional law
advertising
digital media
urban planning
architectural design
Enter your interests separated by commas: legal writing , circuits
No suitable career path found based on the given interests.
aids@aids-System-Product-Name:~$